

EDUCATION

- University of California, San Diego;** M.S. Electrical & Computer Engineering Sep 2025 – Present
• **Focus:** Intelligent Systems, Robotics & Control | **Coursework:** Sensing & Estimation in Robotics, Deep Generative Models
- Indian Institute of Information Technology, Jabalpur;** B.Tech Mechanical Engineering 2019 – 2023
• **GPA:** 3.83/4 (Top 5%) | **Coursework:** Human-centric Robotics, Smart Sensors, Industrial IoT, Data Structures & Algorithms, C++

TECHNICAL SKILLS

Programming	C++ (14/17), Python, C, MATLAB, Bash, CMake, GTest/pytest, Data Structures & Algorithms, Design Patterns
Robotics	ROS/ROS2, DDS, SLAM, Sensor Fusion, Kalman Filter, Motion Planning, Force Control, Sim-to-Real, HIL Testing
AI & Vision	Computer Vision, Deep Learning, Reinforcement Learning, VLMs, Agentic AI, PyTorch, TensorFlow, OpenCV, CUDA
Software	Linux, Real-Time Systems, Multithreading, Docker, CI/CD, Git, Eigen, NVIDIA Isaac Sim/Lab, Gazebo, MuJoCo
Hardware	Humanoids, Quadrupeds, Cobots, Embedded Systems, Jetson Orin, ARM64, 2D/3D LiDAR, Intel RealSense, IMU

PROFESSIONAL EXPERIENCE

Amazon Robotics, Software Development Engineer Intern Jun 2026 – Sep 2026

Agentic Robotics Orchestration

- Prototyped **agentic orchestration** on robot hardware, with a **vision-language model (VLM)** turning commands into executed skills.
- Designed and shipped a **production C++ library** to validate, dispatch, and track robot skills as the shared skill-execution layer.
- Built **non-blocking ROS 2 action/service dispatch** in **thread-safe, RAII, exception-safe C++** behind a middleware-agnostic API.

Reliability, Testing & HIL

- Developed a **fault-injection test harness** over **DDS** that caught a process-crash bug pre-release, improving on-robot reliability.
- Brought up a **Hardware-in-the-Loop (HIL)** simulation bench on ARM64; root-caused **GPU contention** breaking real-time control.
- Enabled research teams to **define skills in JSON** and promote them through validation, HIL testing, and hardware deployment.

Addverb Robotics, Robotics Engineer

Aug 2023 – Jul 2025

Perception, SLAM & Deep Learning

- Lead architect of **object recognition** and **6-DoF pose estimation** modules, designed and deployed from scratch.
- Designed a real-time **ORB-SLAM health indicator** for rapid SLAM diagnostics, cutting debug time by **40%**.
- Engineered a **self-calibrating camera system** that held **± 2 mm accuracy** over long durations by correcting for structural droop.
- Built a **Kalman filter-based sensor fusion** module bridging localization gaps between barcodes, easing brownfield deployment.
- Applied **semantic segmentation** to classify SKUs, chargers, and conveyors for improved robotic perception.
- Linked **scene understanding with error logs** to generate context-aware diagnostics for rare failures, cutting debugging time by **45%**.

Systems & Deployment

- **Dockerized** the ROS-based robotics stack into modular containers with **CI/CD pipelines** that auto-deploy images to robots on push.
- Led **on-site deployments** for **Siemens, ISITEC, Adidas**; integrated Grafana monitoring and time-series root-cause diagnostics.

RESEARCH EXPERIENCE

Student Researcher, Advisor: Prof. Michael Yip

Sep 2025 – Present

Advanced Robotics and Controls Lab (ARClab), UC San Diego

- Implementing a **teacher-student framework** for **zero-shot sim-to-real transfer** in multi-fingered dexterous manipulation.
- Architected the full **NVIDIA Isaac Sim** environment: MDP, action/observation spaces, and reward shaping for stable **RL policies**.
- Developed **density-based curriculum learning** with a safety curriculum for **collision-aware grasping** in RL training.

Research Assistant, Neuro-Biohybrid Systems

Aug 2025 – Oct 2025

Sanford Consortium for Regenerative Medicine, UC San Diego

- Stood up full **quadruped robot** infrastructure: teleoperation, sensor data collection, and **ROS-based data pipeline**.
- Built a **data serialization layer** packaging locomotion telemetry into structured packets for biological neural network training.

Research Intern

Jan 2023 – Jul 2023

Defence Research and Development Organisation (DRDO)

- Researched CNN-based **infrared small target detection (IRSTD)** under low SNR; built annotated IR datasets and **DL pipelines**.

HONORS & ACHIEVEMENTS

- **Published utility patent** for the IoT-Enabled 3D Spatial Monitoring Device 2023
- **Gold Medalist**, IIIT Jabalpur — best cross-disciplinary project among 400+ students 2023
- **International Grant Awardee**, TIGC La Trobe University — Top 10 of 200+ teams across Asia 2022
- **National Winner**, L&T Techgium Hackathon — 1st prize of \$12,000, outperforming 30k+ students from 450 universities 2022